JENNIFER APONTE

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Data engineer with an M.S. in Data Science and four years building and operating the pipelines and warehouse models behind pricing decisions at Hilton. Own production dbt models on Snowflake under version control and peer review, with automated tests and recurring audits that surface defects at the source. Built and operated the production data pipeline feeding a deep learning pricing model covering function space across the global Hilton portfolio, and have designed the schemas that replaced manual spreadsheet workflows with systems other teams depend on.

TECHNICAL SKILLS

Languages: Python (pandas, NumPy, PySpark), SQL (advanced: CTEs, window functions, query optimization, semi-structured data), Bash, R

Pipelines and Orchestration: Apache Airflow (dynamic task mapping, retries, idempotent loads), dbt (models, tests, sources, documentation), ETL and ELT design, incremental and snapshot-based ingestion, data quality testing and reconciliation

Warehousing and Modeling: Snowflake, DuckDB, BigQuery, Redshift, Databricks, Spark, dimensional modeling, schema and grain design, semantic and metric layers, Parquet and columnar storage

Cloud and Infrastructure: AWS, Azure, Google Cloud Platform, Docker, Git and GitHub, CI workflows, peer code review

Modeling and Statistics: Forecasting and backtesting, experiment design, causal inference and quasi-experimental design, scikit-learn, PyTorch, statsmodels

PROFESSIONAL EXPERIENCE

Data Analyst, Revenue Management | Hilton | Frisco, TX *Nov 2024 - Present*

- Write and maintain production dbt models on Snowflake under version control with peer code review, including automated uniqueness and referential integrity tests that catch defects before they reach downstream consumers.
- Developed and operated the production data pipeline feeding that model, owning ingestion, transformation, and the scheduled refresh it depended on.
- Built the training dataset and feature set for a deep learning pricing model covering function space (conference and meeting rooms) across the global Hilton portfolio.
- Designed and own the data model behind the team quality control system, replacing a manual Excel workflow with a normalized schema that standardized definitions and enabled automated failure surfacing, database ingestion, and rework tracking.
- Engineered Python and SQL automation that cut a manual validation process from over 4 hours to 10 to 20 minutes per property, an 80 to 90 percent reduction that absorbed growing volume without added headcount.
- Build and maintain the governed metric definitions, semantic layer, and documentation that make reporting trustworthy, so metric meaning is discoverable and consistent rather than living in individual spreadsheets.
- Use AI coding assistants daily for query generation, code review, debugging, and documentation, with a verification practice that treats generated output as unvalidated until reconciled against source row counts and known totals.
- Present findings, assumptions, and limitations to non-technical partners as decision-ready recommendations, and document methods so the work is reproducible by others.

Data Analyst, Sales Support | Hilton | Frisco, TX *Oct 2023 - Nov 2024*

- Consolidated fragmented pipeline, revenue, and account data from multiple source systems into a single source of truth, standardizing definitions across teams reporting conflicting numbers from the same underlying data; promoted into Revenue Management after 13 months.

Business Operations Processor | Corebridge Financial | Remote *Jun 2022 - May 2023*

- Reconciled high-volume financial transactions against source systems in a regulated environment, querying account and transaction records in SQL and escalating recurring discrepancies for root-cause fixes.

Project Coordinator | Texas Health and Human Services | Austin, TX *Jun 2020 - Jul 2021*

- Coordinated project workflows across state health program teams and began working in SQL to pull and validate program data.

PROJECTS

Short-Term Rental Pricing and Demand Pipeline | github.com/AponteOJennifer/str-pipeline *Airflow, dbt, DuckDB*

- Built an Airflow pipeline ingesting Inside Airbnb snapshots into a DuckDB warehouse with idempotent delete-then-insert loads, and modeled 146,490 listing-months in dbt across staging, intermediate, and mart layers with 24 automated tests.
- Designed data quality models that flag defective source snapshots automatically, catching one scrape missing 2,533 listings and another containing no prices at all, both of which pass a row-count check unnoticed.
- Built and backtested a weekly demand forecast across 12 rolling origins at a 12-week horizon: 7.83 percent MAPE, 34 percent better than a seasonal baseline, with empirical prediction intervals and forecasts keyed on training vintage so accuracy stays auditable.

EDUCATION

Master of Science, Data Science | New Jersey Institute of Technology *Conferred August 2026*

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Applied Statistics, Data Mining, Big Data Analytics

Bachelor of Science, Psychology | University of Houston *May 2020*